



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Fast-Track Innovations with Agile Design Thinking and Generative AI (GenAI)

TGS Ref No: TGS-2024049781

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 4.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
3.0	1 January 2026	Legacy master trainer deck and lesson plan (v3) — 4 topics, exercise-based delivery.	Dr. Alfred Ang
4.0	17 August 2026	Major revision. Content beefed up from Thoughtworks, Adaptovate, BMC, PMI, UX Magazine, The Design Gym, PremierAgile and Startup Frontier. Twelve real-world case-study activities (Airbnb, Netflix/Blockbuster, GE Adventure Series, IDEO, Spotify, IBM, DBS, Boeing 737 MAX) replace the generic exercises, each with scenario, discussion questions and debrief. Design Thinking Studio and Padlet Classroom Board adopted as the class ed-tools.	Dr. Alfred Ang

TABLE OF CONTENTS

Course Information.....	4
Learning Outcomes.....	4
Skills Framework Alignment.....	4
Teaching and Learning Methods.....	5
Learning Resources.....	5
Assessment.....	5
Course Schedule.....	6
Day 1 — Foundations, Problem Framing and Ideation with GenAI.....	6
Day 2 — Agile Delivery, Scaling Innovation and Assessment.....	7
Assessment Session (conducted separately, after the 16 instructional hours).....	8
Activity Reference (aligned to topics and learning outcomes).....	8
Detailed Activity Schedule.....	9

Course Information

Course Title	Fast-Track Innovations with Agile Design Thinking and Generative AI (GenAI)
WSQ Course Reference	TGS-2024049781
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 instructional hours), plus a separate 2-hour assessment session
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks counted within training time)
Mode	Instructor-led, case-study based, with collaborative online ed-tools
Skills Framework	Design Thinking Practice (ICT-ACE-5014-1.1)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Integrate design thinking methodologies and agile principles to drive organisational innovation using generative AI.
- LO2: Synthesise stakeholder inputs to uncover end-user needs for successful innovation.
- LO3: Lead design thinking projects using project management tools and techniques to enhance organisational performance.
- LO4: Develop strategies for agile design thinking to enhance product and service innovation.

Skills Framework Alignment

TSC Title: Design Thinking Practice · TSC Code: ICT-ACE-5014-1.1

Abilities:

- A1 – Integrate design thinking methodologies into processes to drive innovation across the organisation
- A2 – Develop strategies to proliferate design thinking across the organisation
- A3 – Synthesise information from different sources and stakeholders in order to fully understand the needs of end users
- A4 – Drive the development of new strategies to enhance products and/or services for the organisation
- A5 – Engage stakeholders during the design thinking process to uncover the motivations behind their actions and behaviours
- A6 – Cultivate design thinking as a viable tool and methodology to foster new innovations for the organisation
- A7 – Lead design thinking projects across the organisation

Knowledge:

- K1 – Latest trends in design thinking
- K2 – Concept of innovation management
- K3 – Drivers of organisational growth and success
- K4 – Concept and principles of resource management
- K5 – Project management tools and techniques

Teaching and Learning Methods

- Trainer-led concept delivery using a highly visual slide deck (diagrams, comparison tables and frameworks rather than bullet walls).
- Twelve real-world case-study activities, each carrying a documented case, a workplace scenario, five discussion questions and a trainer debrief.
- Collaborative group work in groups of 3–5 using two online ed-tools: the Design Thinking Studio (5-stage workspace) and the Padlet Classroom Board.
- Guided use of generative AI tools (ChatGPT / Microsoft Copilot) with explicit critique of AI output against evidence.
- Cross-group peer review — groups critique each other's output and defend their reasoning.
- Continuous formative feedback from the trainer during each activity, and a structured debrief after each one.

Learning Resources

- Trainer Slides (PPT/PDF), Learner Guide (PDF) and this Lesson Plan (PDF) — downloadable from the LMS at <https://lms-tms.tertiaryinfotech.com>.
- Twelve individual activity briefs (PDF), one per activity, each containing the case, the scenario, the discussion questions, the step-by-step instructions and the debrief.
- Design Thinking Studio — <https://alfredang.github.io/designthinking/>
- Padlet Classroom Board — <https://alfredang.github.io/padlet/>
- A generative AI assistant (ChatGPT or Microsoft Copilot) and a laptop with a modern web browser.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 0.5 hour, open book.
- Case Study (CS) — a scenario-based innovation case, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide, activity briefs and approved materials only.
- The final assessment is conducted as a separate session at the end of Day 2, outside the 16 instructional hours: WA (0.5 h) then CS (1 h).
- Learners are assessed as Competent (C) or Not Yet Competent (NYC) on each instrument.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Foundations, Problem Framing and Ideation with GenAI

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, trainer and learner introductions, ground rules, learning outcomes, course outline and mandatory SSG digital attendance (AM). Introduction to the two course ed-tools: Design Thinking Studio and the Padlet Classroom Board.
10:00–11:00	60 min	Topic 1 — Foundations of Design Thinking, Agile and Generative AI: what design thinking is; the three mindsets (Design Thinking, Lean, Agile) and the question each answers; the Double Diamond; the five stages; wicked problems.
11:00–11:15	15 min	Tea break
11:15–12:00	45 min	Topic 1 continued — Product development through design thinking and agile; how generative AI enhances the loop; where AI must not replace human judgement; latest trends in innovation.
12:00–13:00	60 min	Hands-on case study: Activity 1: Airbnb — Diagnosing Why a Method Alone Does Not Save a Business
13:00–14:00	60 min	Lunch break
14:00–14:15	15 min	Mandatory SSG digital attendance (PM)
14:15–15:30	75 min	Hands-on case studies: Activity 2: Netflix vs Blockbuster — Innovation as an Operating System; Activity 3: Positioning GenAI in the Innovation Loop — Where It Helps and Where It Lies
15:30–15:45	15 min	Tea break
15:45–16:45	60 min	Topic 2 — Problem Framing and Ideation: framing the design challenge; the Problem-Assumption Model; problem statement vs Point of View vs How Might We; the altitude test; empathy maps and persona mapping with GenAI; ideation techniques.
16:45–18:15	90 min	Hands-on case studies: Activity 4: GE Adventure Series —

		Reframing the Problem Instead of Fixing the Machine; Activity 5: IDEO Shopping Cart — Volume, Then Judgement
18:15–18:30	15 min	Day 1 recap, Q&A and close

Total training time: 480 minutes (8 hours).

Day 2 — Agile Delivery, Scaling Innovation and Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory SSG digital attendance (AM)
9:45–10:45	60 min	Topic 2 concluded — prototyping with AI-driven tools and defining a falsifiable test plan. Hands-on case study: Activity 6: Rapid Prototyping with AI — From Concept to Testable Artefact in One Hour
10:45–11:00	15 min	Tea break
11:00–12:00	60 min	Topic 3 — Agile Development and AI for Rapid Solution Delivery: being agile vs doing agile; the Scrum loop and the three accountabilities; dual-track agile; the mini-waterfall anti-pattern; Epic → Feature → Hill → Story → Task.
12:00–13:00	60 min	Hands-on case study: Activity 7: Spotify — Reading a Scaling Model Honestly
13:00–14:00	60 min	Lunch break
14:00–14:15	15 min	Mandatory SSG digital attendance (PM)
14:15–15:30	75 min	Topic 3 continued — user stories and the 3 C's, the three-cake release model, story points and Planning Poker, GenAI across the sprint. Hands-on case studies: Activity 8: From Hills to Backlog — Converting Ideas into Stories with GenAI; Activity 9: Running Dual-Track Agile — A Sprint Simulation with AI Assistance
15:30–15:45	15 min	Tea break
15:45–16:45	60 min	Topic 4 — Scaling and Sustaining Innovations: the three levers that scale innovation; culture and the four principles; stakeholder buy-in; resource management with AI; sensemaking; systems thinking and feedback loops; metrics and KPIs.
16:45–17:45	60 min	Hands-on case studies: Activity

		10: DBS Bank — Scaling Innovation Across 25,000 People; Activity 11: Boeing 737 MAX — Systems Thinking and the Cost of Local Optimisation; Activity 12: Innovation Metrics — Designing KPIs That Change a Decision
17:45–18:00	15 min	Course revision and Q&A
18:00–18:30	30 min	Briefing for Assessment, mandatory TRAQOM survey and Assessment digital attendance. End of the 8 instructional hours.

Total training time: 480 minutes (8 hours).

Assessment Session (conducted separately, after the 16 instructional hours)

The 2-hour assessment is scheduled as a separate session at the end of Day 2 and is NOT counted within the 16 instructional hours. Learners who have met the 75% attendance requirement sit both instruments in the order below.

Duration	Instrument	Details
15 min	Briefing for Assessment	Assessment rules, materials permitted, and the appeal process.
0.5 hour	Written Assessment (WA) — Short-Answer Questions (SAQ)	6 open-ended knowledge questions covering K1–K5. Open book.
1 hour	Case Study (CS)	1 scenario with 4 tasks covering A1–A7. Open book.
15 min	Submission and records	Learners upload answers to the LMS and sign the Assessment Summary Record.

Activity Reference (aligned to topics and learning outcomes)

Topic	Weighting	Activities	Ed-tool
Topic 01: Foundations of Design Thinking, Agile, and Generative AI for Problem-Solving	25%	Activity 1, Activity 2, Activity 3	Padlet
Topic 02: Problem Framing and Ideation: Leveraging Design Thinking and Generative AI	25%	Activity 4, Activity 5, Activity 6	Design Thinking Studio
Topic 03: Agile Development and AI	25%	Activity 7, Activity 8, Activity 9	Design Thinking Studio / Padlet

for Rapid Solution Delivery			
Topic 04: Scaling and Sustaining Innovations with Agile Design Thinking and Generative AI	25%	Activity 10, Activity 11, Activity 12	Design Thinking Studio / Padlet

Detailed Activity Schedule

#	Activity	Duration	Real-world case
1	Airbnb — Diagnosing Why a Method Alone Does Not Save a Business	45 min	Airbnb, 2009...
2	Netflix vs Blockbuster — Innovation as an Operating System	45 min	In 2000 Reed Hastings offered to sell Netflix to Blockbuster for \$50 million...
3	Positioning GenAI in the Innovation Loop — Where It Helps and Where It Lies	40 min	A Singapore bank's innovation team used a large language model to generate customer person...
4	GE Adventure Series — Reframing the Problem Instead of Fixing the Machine	50 min	GE Healthcare's MRI scanners were technically excellent, but designer Doug Dietz watched a...
5	IDEO Shopping Cart — Volume, Then Judgement	50 min	In 1999, ABC's Nightline gave the design firm IDEO five days to redesign the supermarket s...
6	Rapid Prototyping with AI — From Concept to Testable Artefact in One Hour	50 min	A Singapore hospital team needed to test whether a self-service kiosk would cut queue time...
7	Spotify — Reading a Scaling Model Honestly	45 min	In 2012 Henrik Kniberg and Anders Ivarsson published a paper describing how Spotify organi...
8	From Hills to Backlog — Converting Ideas into Stories with GenAI	55 min	IBM runs a two-day Design Thinking and Agile workshop in which Day 1 produces 'Hills' — ou...
9	Running Dual-Track Agile — A Sprint Simulation with AI Assistance	50 min	Marty Cagan describes the most common failure he sees in teams that believe they are agile...
10	DBS Bank — Scaling Innovation Across 25,000 People	50 min	Around 2014 DBS Bank set out to 'Make Banking Joyful' and to become a technology

			company d...
11	Boeing 737 MAX — Systems Thinking and the Cost of Local Optimisation	50 min	Boeing needed a fast, cheap answer to the Airbus A320neo...
12	Innovation Metrics — Designing KPIs That Change a Decision	45 min	Two innovation programmes reported to the same board...